Bags, Vectors & Transformers

Text as Data for Causal Questions · Bonus, Part 12

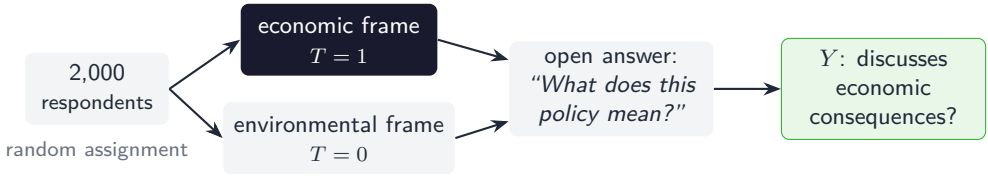
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Three Roles for Text in a Causal Analysis

role	example	main danger
outcome	Does an economic frame make people *write* more about economic consequences?	measurement error, especially if it differs between groups
treatment	Which features of a petition *cause* more signatures?	discovering the "treatment" and testing it on the same data
confounder	Do conflict-framed articles get shared more, *given* their topic?	adjusting for text that contains the treatment, or that the treatment changed

The logic is the usual one (randomize, or adjust for confounders), but the variables are **measured from text by a model**. The model's errors and the researcher's choices become part of the design. The notebook simulates everything, so every estimate can be checked against the truth.

A Framing Experiment with a Text Outcome



True effect in the simulation: 30% $\rightarrow$ 42%, **+12 points** (+16 in the drawn sample). Y is measured with a **dictionary** of economic words, standing in for any measure not built on this data: an existing classifier, a zero-shot LLM.

The complication: some treated respondents *echo* the article ("it was all about costs and money") without discussing consequences themselves. Not economic by the codebook, but full of economic words.

When Measurement Error Biases the Effect

estimate	effect
truth (unknown in practice)	0.161
dictionary, all 2,000	**0.246**

Error rates, from the hand-coded random sample:

	false pos.	false neg.
control	0.12	0.22
treatment	**0.47**	0.31

Non-differential error (same rates in both groups): the effect is shrunk toward zero, by $1 - \text{FPR} - \text{FNR}$. You may miss an effect; you won't invent one.

Differential error (rates depend on the treatment): bias in **either direction**, possibly large. The realistic case with text, because treatments change *language*: vocabulary, length, tone.

Validate per group, not just overall.

Fixing It: Design-Based Supervised Learning

Hand-code a **random** sample with known probability π (here 15%). For every document:

$$\tilde{Y}_i = \hat{Y}_i + \frac{R_i}{\pi_i}\left(Y_i - \hat{Y}_i\right)$$

$\hat{Y}$: automatic measure; $R_i = 1$ if hand-coded. The weighted corrections cancel the measure's errors *on average*, whatever they look like. Then estimate the effect on $\tilde{Y}$. (Egami et al., 2023; R package `dsl`)

If $\hat{Y}$ comes from a classifier trained on the hand-coded documents, predict those documents out of sample (cross-fitting).

500 draws of the hand-coding sample

	bias	sd	coverage
dictionary	.085	.000	0%
gold only (≈300)	.000	.049	97%
DSL, dictionary	.001	.053	96%
DSL, trained classifier*	.006	**.018**	

*100 draws. Coverage: share of 95% intervals containing the truth.

The dictionary alone is **confidently wrong**. DSL is unbiased; with a *good* measure it's also far more precise than gold only.

Don't Discover and Test on the Same Data

With text, the measure is often *discovered* from the data: a dictionary built by reading answers, a topic picked from a topic model, a codebook tuned on the sample. Testing it on the **same** data tests a measure tailored to this sample's quirks.

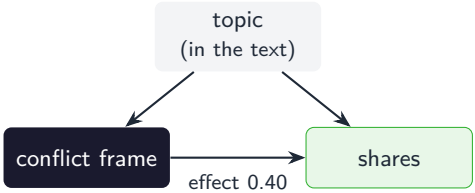
Split sample (Egami et al., 2022): discover on one part, test on the other, as if pre-registering the measure.

An experiment with no effect at all

Pick the 5 words most over-represented among the treated, then test whether answers with them differ between groups. 200 random splits:

	"significant"
tested on the same half	**28%**
tested on the other half	4%
expected under no effect	5%

Text as a Confounder



adjusted for	estimate
nothing	0.78
true topic	0.44
NMF topics, conflict words removed	0.44
NMF topics, all words	0.41

Political articles are more often conflict-framed *and* more shared. Estimate the topic from the text (topic model, NMF) and adjust for it.

Trap 1: if the text representation also captures the *treatment* (the conflict words), adjusting for it absorbs part of the effect. Measure confounders without the treatment.

Trap 2: don't adjust for text the treatment *changed*. A sensational headline caused by the frame: total effect ≈0.59 in the sample; adjusting for the headline gives 0.43.

Text as Treatment, and a Checklist

Text as treatment

- **randomize texts**: versions that differ in one feature (survey experiments, A/B tests). Cleanest; the hard part is changing *only* that feature
- **discover treatments** in observational texts (Fong & Grimmer, 2016), with a split sample; texts bundle many features that are hard to separate

Checklist

- outcome, treatment or confounder? Each has its own pitfalls
- validate your measure **per group**: differential error is the dangerous kind
- hand-code a **random sample with known probabilities**, and use it in estimation (DSL), not only to report accuracy
- don't discover and test on the same data
- adjusting for text: exclude the treatment; no post-treatment features
- LLM-based measures are classifiers too

In the Notebook

- A simulated framing experiment with open-ended answers
- Non-differential vs. differential measurement error
- Design-based supervised learning, tested over 500 repetitions
- A trained classifier with cross-fitting as a better measure
- Why discovery and testing need separate data
- Adjusting for text as a confounder, and two traps

`bonus/notebooks/12_text_as_data_for_causal_questions_exercises.ipynb` · 5 exercises · no GPU needed